

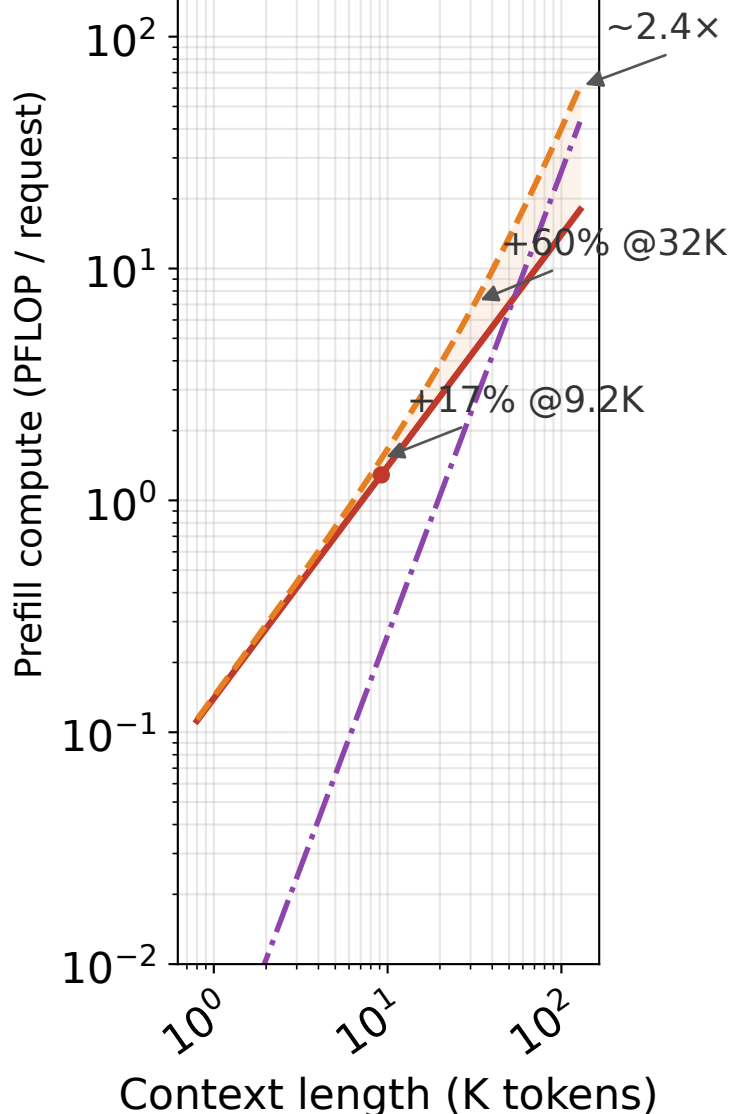


# LEFT: PER REQUEST | RIGHT: PER GENERATED TOKEN

**DO NOT COMPARE Y-AXIS MAGNITUDES DIRECTLY — different units, different baselines**

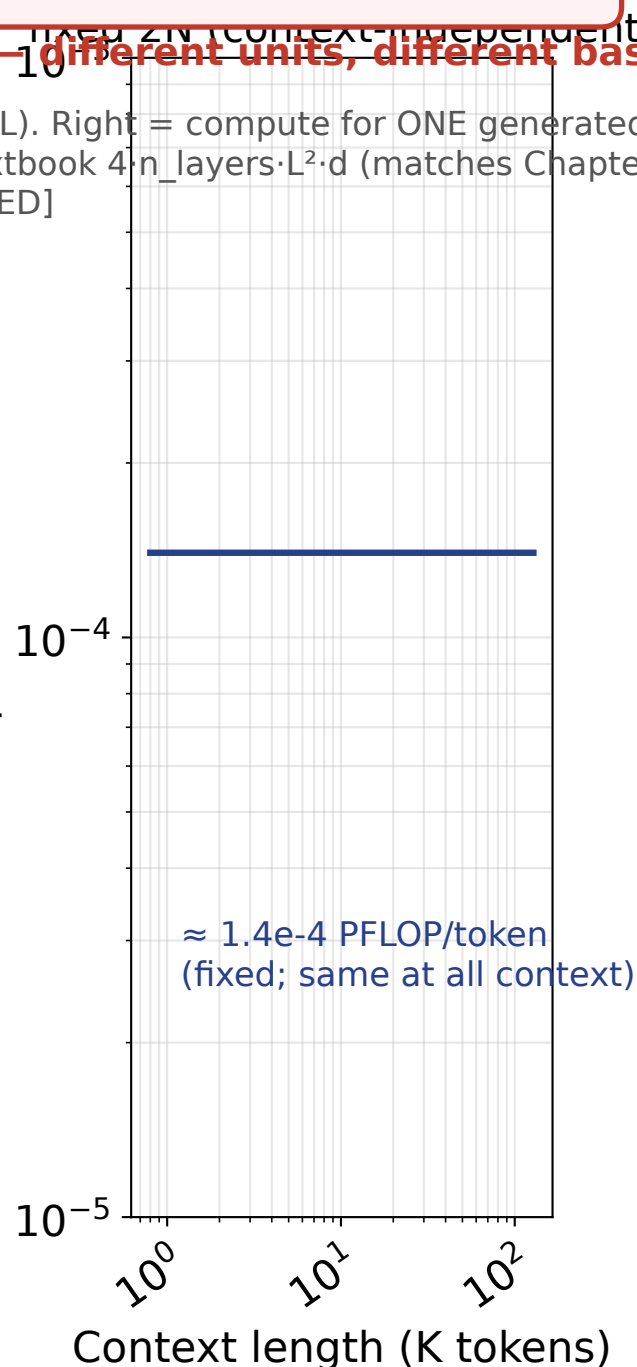
Left = total prefill compute for ONE request (grows with context L). Right = compute for ONE generated token (fixed 2N, context-independent). The quadratic term is the textbook  $4 \cdot n_{\text{layers}} \cdot L^2 \cdot d$  (matches Chapter 8).

[ILLUSTRATIVE][DERIVED]



- linear term ( $2 \cdot N \cdot L$ )
- · - quadratic attention term ( $4 \cdot n_{\text{layers}} \cdot L^2 \cdot d$ )
- - - total = linear + quadratic

Decode compute (PFLOP / token)



$\approx 1.4 \times 10^{-4}$  PFLOP/token  
(fixed; same at all context)